

Global Supply Chain Intelligence

Portfolio Project Documentation & GitHub README Guide



Project Overview

This project addresses the complexities of global logistics by analyzing a dataset of **180,519 transactions**. The goal was to build a robust pipeline that handles high-volume data cleaning, relational database migration, and multi-layered visualization. By uncovering patterns in shipping delays, fraudulent transactions, and profit leakage, this analysis provides a 360-degree view of supply chain health.



Tech Stack

- **Languages:** Python (Pandas, NumPy), SQL (PostgreSQL)
- **Data Engineering:** SQLAlchemy (High-speed Batch Processing)
- **Visualization:** Tableau Desktop, Seaborn, Matplotlib
- **Tools:** VS Code, pgAdmin 4, Git/GitHub, Virtual Environments



Key Features & Analysis

1. Data Engineering & ETL (Python)

- **Encoding & Normalization:** Resolved complex encoding conflicts and standardized 60+ inconsistent column names for database compatibility.
- **Optimized Migration:** Implemented a `chunksize=10000` upload strategy to migrate 180k rows into PostgreSQL efficiently.
- **Feature Engineering:** Developed metrics like `is_late` (binary indicator) and `profit_margin_pct` to assess fulfillment health.

2. Relational Database Design (PostgreSQL)

Automated the transfer of processed DataFrames into a structured PostgreSQL database. Advanced SQL analysis includes:

- **Logistics BI:** Late delivery rates segmented by Market and Shipping Mode.
- **Financial Auditing:** Identifying high-revenue categories with negative profit margins.
- **Strategic Ranking:** Utilizing Window Functions (`RANK()` , `PARTITION BY`) for category-level product performance.

3. Business Intelligence (Tableau)

Logistics Risk Map: A geographic heatmap identifying high-delay clusters (e.g., LATAM and Africa).

Product Profitability Matrix: A quadrant analysis categorizing products into "Stars," "Loss Leaders," and "Niche" items.

Shipping Variance: Box-and-Whisker plots showing the variance between 'Scheduled' and 'Real' shipping days.

Repository Structure

```
├─ notebooks/      # supply_chain_analysis.ipynb (Cleaning & ETL)
├─ sql_queries/    # analysis_queries.sql (Easy, Medium, Hard scripts)
├─ tableau/        # .twbx workbook and dashboard screenshots
├─ requirements.txt # List of Python dependencies
├─ .gitignore       # Configuration to exclude venv/ and caches
└─ README.md       # Full project documentation
```

How to Run

1. **Clone:** `git clone <repo-url>`

2. **Setup Environment:**

```
python -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\ctivate
pip install -r requirements.txt
```

3. **Run ETL Pipeline:** Execute the Jupyter notebook to push data to PostgreSQL.

4. **SQL Analysis:** Run the scripts in `/sql_queries` via pgAdmin.